ECE 285: preliminary dataset and generative-model experiments

Ricardo, Beihao and Varad 6 November 2026

1. What we are trying to learn

The immediate question is whether a conditional generative model can propose new, low-energy arrangements of lithium vacancies efficiently. We also test whether a learned coordinate predictor can make subsequent geometry relaxation cheaper. These are related but different questions: choosing which Li atoms remain and finding where the remaining atoms settle need separate validation. Our conditional variational autoencoder (CVAE) samples arrangements at a requested lithium fraction and energy class.

Lithium cobalt oxide ($LiCoO_2$, LCO) is our first material. Cathode recycling motivates the work. Removing lithium changes both its spatial ordering and the response of the cobalt–oxygen framework. A computational model that handles these states reliably could help characterize recycled electrodes. The present calculations contain bulk cathode structures, without water, electrolyte or a lithium-source reaction. They do not directly predict a hydrothermal reaction rate or optimal reactor conditions.

Three computational tools have different roles here. **DFT**, density functional theory, supplies the electronic-structure reference energies, forces and stresses in the historical and Hu archives. Those references depend on their chosen numerical and physical settings. **MLIPs**, machine-learned interatomic potentials, predict the same quantities much more cheaply and act as frozen scoring models in these preliminary experiments. **MD**, molecular dynamics, evolves atomic positions using those forces. Short MD trajectories provide thermal geometries; equilibrium thermodynamic predictions require additional sampling and validation. No new DFT calculations were submitted.

Main observations. The vacancy CVAE has not yet beaten a strong local mutation baseline. Its average advantage over uniform sampling is small and seed-dependent. Coordinate prediction is more promising as a relaxation starting point, but deterministic regression performs similarly to the geometry CVAE. None of the decoded geometries is already force-converged. The study can focus on why conditioning succeeds or fails and how much relaxation work the generated starting points save.

Completed component	Size
Symmetry-unique within-parent catalog, five cells	32,855
Frozen-model static energy/force/stress labels	164,275
Distinct main-cell mask lineages selected for relaxation	1,000
Fixed-cell / variable-cell model–parent attempts	5,000 / 5,000
Fresh candidate symmetry orbits, scored by all five models	54,752
Fresh model–candidate evaluations	273,760
Prior-decoded geometry comparisons	128 masks, 5 starts, 5 models
Matched UMA relaxations of decoded starts	32 masks, 3 starts
Thermal production dataset	200 trajectories; 4000 exported frames

The models are UMA-S-1.2.1 (`omat`), MACE-MH-1 (`omat_pbe`), SevenNet-Omni (`mpa`), CHGNet-PES-MatPES-PBE-1M-2026.9 and M3GNet-PES-MatPES-PBE-2025.2. Checkpoint hashes, environments and FP32 settings are recorded; these labels do not imply equivalent DFT reference Hamiltonians.

2. Structures, symmetry and independent validation units

The main parent is O3 layered LCO with 48 possible Li sites, 48 Co and 96 O. For occupancy vector $o = (o_1, \ldots, o_{48})$, $o_i = 1$ means Li is present and $o_i = 0$ means a vacancy. Thus $n = \sum_i o_i$, the Li fraction is $x = n/48$, and the atom count is $A = 144 + n$. All compositions from empty to fully occupied are represented in the catalog. The initial sampler comparison uses $n = 12, 24, 36$. Its resolution in composition is 1/48; it does not require a 144-Li parent.

Parent	Li sites	Distinct catalog masks
Main O3 pilot parent	48	21,294
Hu O3	12	87
Historical O3	24	2,138
Compact O3	36	3,968
Alternative compact O3	48	5,368

The 87 Hu masks are the exhaustive symmetry orbits of all $2^{12} = 4096$ binary masks for that particular periodic parent. The other larger catalogs are samples, not exhaustive configuration enumerations. Cell shape changes the retained symmetries, so the two 48-site parents are different namespaces. The Hu count is exact under its 144 retained site permutations, constructed at `symprec`=0.001 Å; the original parent has a maximum symmetry-mapping residual of 0.000264 Å. Its atomic-coordinate symmetry is therefore approximate.

A **symmetry orbit** contains masks transformed into each other by species-preserving translations and point symmetries of the periodic parent. We assign the smallest binary vacancy string over that orbit as its canonical key. For the main parent there are 192 retained site permutations. This removes symmetry copies of a vacancy arrangement; it does not identify all relaxed or thermally distorted geometries with that arrangement as identical structures.

A four-site ring illustrates the distinction. At two occupied sites there are six raw masks: four adjacent pairs and two opposite pairs. There are only two rotation-unique orbits. Sampling the six raw masks uniformly gives orbit probabilities 2/3 and 1/3, not 1/2 each. Our uniform baseline similarly samples raw fixed-count masks and then canonicalizes them.

The train/validation/test split is frozen *before* symmetry augmentation. All model labels, relaxation frames and temperatures descended from a mask stay in its split. A trajectory with 100 frames is one correlated lineage, not 100 independent configurations. Exported trajectories are capped and weighted to prevent long runs from dominating. Sparse end-member groups are diagnostic.

Two generalization questions must remain separate. Withholding interior Li counts tests **composition interpolation**: those counts lie within the training composition range. Training on another supercell and testing a smaller cell would test size transfer, which has not been demonstrated here. Primitive periodic aliases between different parents need checking before a mixed-size split can be called independent. The dense models in this package accept one 48-site parent; auxiliary cells are presently diagnostic data.

The main mask catalog and these splits are in `data/parents/`, `data/mask_manifest.csv` and `splits/lineage_splits.csv`. The independent mask audit checks all Hu masks and forward symmetry operations.

3. What the CVAE condition and loss actually mean

Within a teacher model and Li count, let $e = E/A$ be energy per atom. From the m training values only, define $q_{0.2}$ as sorted value number $\lceil 0.2m \rceil$, including ties. The binary energy label is $b = \mathbf{1}[e \le q_{0.2}]$ and the condition is $c = (n/48, b)$. The label $b = 0$ means the remaining approximately 80%, not the highest 20%. Defining the class separately by composition avoids treating removal of an atom as though it were a same-composition energy improvement.

The encoder $q_\phi(z \mid o, c)$ predicts a mean μ and diagonal variance σ^2 for an eight-component latent vector z. During training, $z = \mu + \sigma \odot \epsilon$, with $\epsilon \sim \mathcal{N}(0, I)$; the decoder predicts one occupancy logit ℓ_i per site. ϕ and θ denote encoder and decoder weights. A logit gives probability $p_i = (1 + \exp(-\ell_i))^{-1}$. The loss per mask is

$$
\begin{aligned}
L_{\text{rec}} &= -\sum_{i=1}^{48} \{o_i \log p_i + (1 - o_i) \log(1 - p_i)\}, \\
L_{\text{KL}} &= \tfrac{1}{2} \sum_j (\mu_j^2 + \sigma_j^2 - 1 - \log \sigma_j^2), \\
L &= \mathbb{E}_{q_\phi}[L_{\text{rec}}] + \beta L_{\text{KL}}.
\end{aligned}
$$

Reconstruction rewards explaining an observed mask. KL divergence penalizes departure from the standard-normal prior used to generate new masks. At $\beta = 1$ this is the negative evidence lower bound (ELBO), up to the usual sampling estimate. Lower β changes the tradeoff; it is not the same likelihood objective. We warm β over 20 epochs. Every run, including $\beta = 0.1$, selects checkpoints using ordinary validation negative ELBO. The derivation follows Kingma and Welling, Sec. 2.2, Eqs. (1)–(3), and the conditional construction of Sohn et al., Sec. 3 (references on the last page).

At generation time, sample $z \sim \mathcal{N}(0, I)$ and request $b = 1$. That request is a learned preference, not a physical constraint or energy gradient. It cannot guarantee energy below a threshold, a value below the training minimum, or the true ground state.

We separately enforce $\sum_i o_i = n$. Top-count decoding selects the n largest logits: the constrained most-probable mask for those logits. Dynamic programming instead samples, for fixed z, c, the exact distribution

$$
p(o \mid z, c, \textstyle\sum_i o_i = n) = \frac{\exp(\sum_i \ell_i o_i)}{Z_n(\ell)} \, \mathbf{1}[\textstyle\sum_i o_i = n].
$$

Z_n sums the numerator over fixed-count masks. This exact per-logit sampler is not Gumbel-top-n; drawing the original latent prior first is also distinct from globally conditioning the original joint model on atom count. Training ELBO evaluates the unprojected Bernoulli model, not this count-conditioned loss.

Composition-only CVAE and unconditional VAE controls use the same latent dimension, hidden-layer widths and count rule; conditioning changes the first layer's parameter count. The VAE receives its count only through decoding. Thus 100% projected count validity is imposed, rather than learned success. Raw expected-count error and Bernoulli/threshold-count errors are reported too.

4. Fresh scoring: quality, locality and seed sensitivity

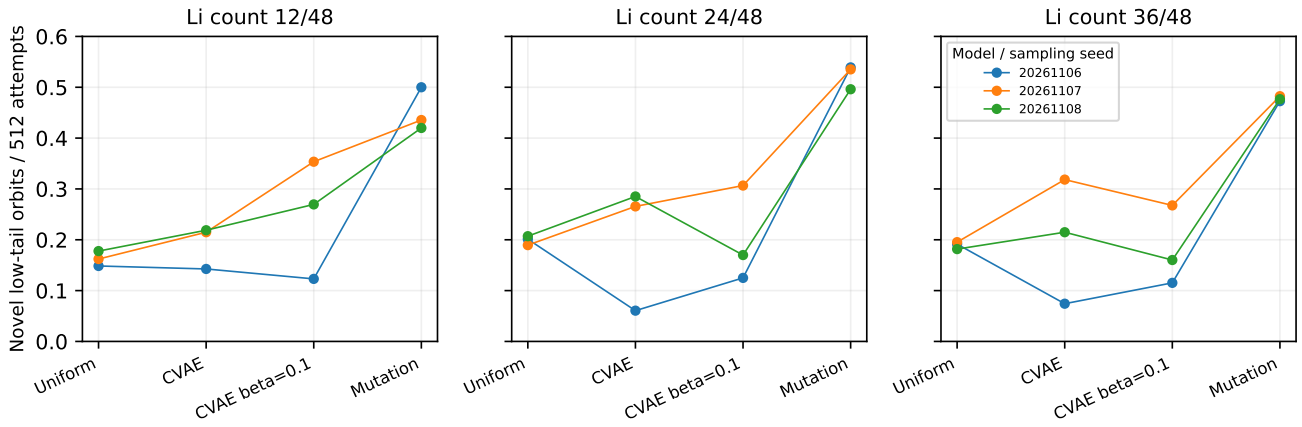
We froze 16 fitted models before scoring 56,832 proposal attempts. They produced 54,752 new symmetry orbits, absent from the complete catalog. Every new orbit was evaluated at the same deletion geometry by all five frozen MLIPs. Each method/composition/seed has 512 attempts, including repeated proposals. The main score is

$$Y = \frac{\#\{\text{distinct novel orbits with } e \le q_{0.2}\}}{512}.$$

Thresholds come from each assessment teacher's own training labels. Energies are compared only within teacher and composition. We also retain minimum, median, lower-tail energy, duplicate rate and standalone unique scoring cost. Actual shared-union cost is stored separately to avoid charging cached results to whichever method appears first.

UMA assessment, mean over 3 counts × 3 seeds	Yield	Nearest train swaps
Uniform fixed count	18.4%	3.67
Standard UMA CVAE	19.9%	3.67
UMA CVAE, $\beta = 0.1$	21.0%	3.67
Low-training-energy mask mutation	48.4%	1.98

Mutation exchanges one to three Li/vacancy pairs from an observed low-energy training mask. Its advantage comes with more local exploration. Distances above are exact minima over training masks and all parent symmetries; Hamming distance divided by two is the minimum number of fixed-composition swaps. A mask can be novel but only one swap from training data.



These averages conceal substantial variability. For one standard-CVAE seed, the three UMA yields were about 14.3%, 6.1% and 7.4%; another gave 21.5%, 26.6% and 31.8%. Three seeds give limited evidence about training variability. Sampling noise is separate: 512 draws from one fit are not 512 independently trained models, and unique-orbit counts couple repeated draws. The held-composition CVAE achieved 11.4% against 19.9% uniform under UMA, using separately labeled withheld-catalog assessment thresholds. This is an informative failure. That held-composition average covers six counts from one CVAE fit and one uniform sampling seed, rather than the primary experiment's three-seed design.

Standard fits have very small latent KL; reducing β increased latent use and improved posterior reconstruction, but did not solve prior-generation quality. Low versus not-low conditions shift paired UMA energies by about -0.475 meV/atom on average. Conditioning has an effect, without producing a strong sampler. Detailed 555-row metrics and seed-specific results accompany the figure. Agreement between MLIPs is consistency evidence, not DFT accuracy.

5. Geometry prediction: useful starting points, not optimized outputs

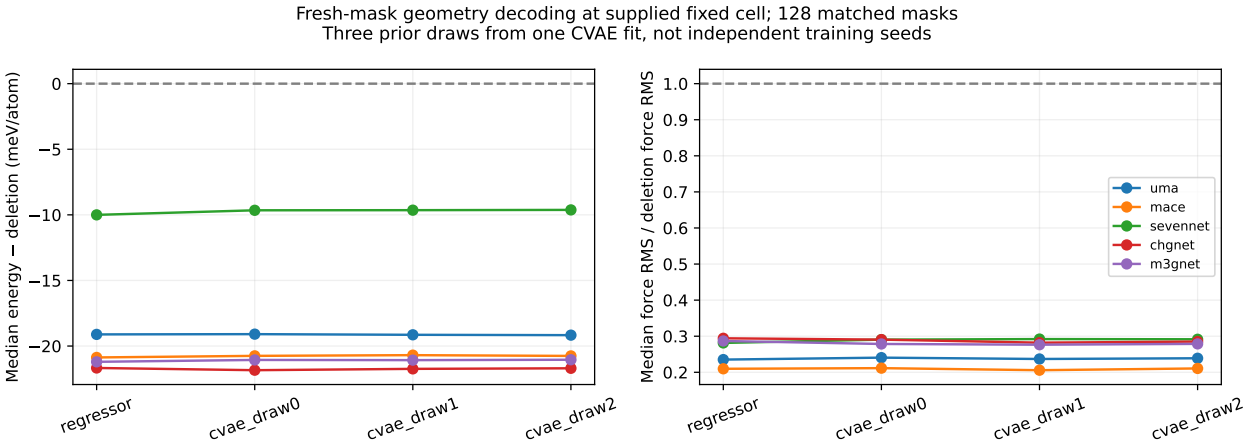
A vacancy mask alone describes atoms at pristine positions. Removing Li can leave nonzero forces. A displacement u_a describes how a present atom a moves from its parent site. We map pristine fractional sites through the frame cell H, subtract the global mass-weighted translation and apply the periodic minimum-image convention. Cell deformation $H_0^{-1}H$ is stored separately. This removes global motion and affine strain from internal displacement.

The prototype factors the proposal into a mask model and a coordinate model:

$$p(o, u \mid c_o, c_u, H) = p(o \mid c_o)\, p(u \mid o, c_u, H).$$

It is a two-stage prototype, not one jointly trained end-to-end network. Only the original fixed cell was used; another supplied cell is untested. The coordinate model uses 1,000 parent masks and 6,000 capped UMA fixed-cell frames. Its context contains occupancy, Li fraction, geometry cohort, supplied cell deformation, and four scalar properties: energy per atom, force component RMS, configurational pressure and deviatoric-stress norm. The last two describe isotropic compression and non-isotropic stress. Force RMS is $[\sum_a \|F_a\|^2/(3A)]^{1/2}$ in eV/Å. Pressure is $-\mathrm{Tr}(\sigma)/3$; deviatoric stress is $\sigma_{\mathrm{dev}} = \sigma - \mathrm{Tr}(\sigma)I/3$. Its Frobenius norm and pressure are the two rotationally invariant stress summaries.

Energy/atom and these summaries are standardized using training frames only, within composition and cohort. The force/stress transforms are $\log(1 + F_{\mathrm{RMS}}/(1\,\mathrm{eV/Å}))$ and $\log(1 + \|\sigma_{\mathrm{dev}}\|_F/(1\,\mathrm{GPa}))$. Desired conditions are actual attained training endpoint tuples, not invented zero-force targets. The coordinate loss is a presence-masked Gaussian reconstruction loss plus KL. No MLIP energy/force-gradient penalty is included. Properties used as inputs are not the same as physical loss weights.



For 128 fresh masks, all learned starts lowered energy under every teacher. UMA CVAE median force RMS was 0.24 times deletion-only RMS; the regressor was 0.235. Across models the first CVAE draw gave ratios 0.21–0.29. **Zero of 128 starts per model met $F_{\max} < 0.05$ eV/Å.** Learned coordinates still require relaxation.

We chose 32 masks before examining decoder energies and used the first CVAE draw, never the best of three. All 96 matched UMA relaxations converged. Median steps were 40 from deletion, 24 from regression and 24 from the CVAE; median paired CVAE saving was 15 steps. Endpoint energies differed from deletion relaxations by at most 0.114 meV/atom. This is not proof of the same basin or ground state. Resident decoder and count-sampler timings are recorded separately; they exclude loading, oracle calls and relaxation. Step savings therefore do not yet establish an end-to-end speedup. Three draws from one fit are one training seed.

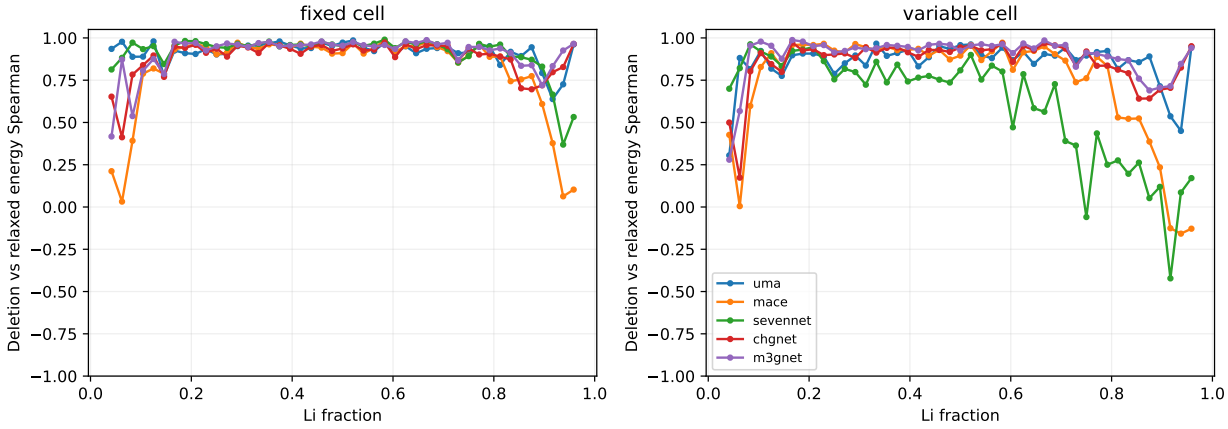
6. Cell relaxation changes the target being learned

Fixed-cell optimization moves atoms while keeping H constant. Variable-cell optimization also changes lattice size and shape at zero external pressure. We use provisional numerical stopping targets of maximum atomic force below 0.05 eV/Å and, for variable cells, maximum absolute principal stress below 0.1 GPa, with a 500-step cap. The stress test uses tensor eigenvalues, not the largest Voigt component. These are explicit pilot tolerances, not universal criteria for thermodynamic precision or DFT accuracy.

Variable-cell optimization was calibrated before production. Fresh FIRE controls with cell scale A converged in 5/25 cases, versus 25/25 with scale $A/4$. Changing the filter scale changes optimizer conditioning, not the physical target. We therefore test physical forces and stress directly, avoiding a scale-dependent extra generalized-force gate. Selected settings are FIRE/$A/4$ for UMA and LBFGS/$A/4$ for the other models. A further 100 confirmations passed.

Model	Fixed converged	Variable converged	Median fixed steps	Variable steps
uma	1000/1000	1000/1000	38	66
mace	1000/1000	1000/1000	36	39
sevennet	1000/1000	1000/1000	31	63
chgnet	1000/1000	999/1000	40	37
m3gnet	1000/1000	998/1000	34	39

The three capped attempts are retained and flagged. The audit independently checks physical endpoint criteria, trace lengths, positive cell volumes and fixed-cell invariance. All 10,000 logical attempts and 109,498 raw snapshots are accounted for. A recovered infrastructure interruption uses the first valid original result or its explicitly linked retry, never the lower-energy attempt.



The paired comparison shows that **deletion energy and relaxed energy are different objectives**. UMA fixed-cell rank correlations at $n = 12, 24, 36$ are 0.902, 0.972 and 0.914, close but not identical. Some variable-cell cases reorder much more strongly. These correlations describe the acquisition-selected 1,000 parents, with only converged endpoints and sufficient plotting support; they are not an unbiased population estimate. Near-degenerate energies and finite stopping tolerances can also change ranks.

If the final objective is relaxed low-energy vacancy arrangements, train and evaluate against relaxed energies. Do not silently use a deletion-energy CVAE as though it optimized the relaxed landscape. Large SevenNet cell/coordinate changes warrant structural inspection before any phase-transition claim.

7. Thermal data and the boundary of physical validation

MD uses $m_a\ddot{r}_a = F_a$ to evolve atoms. An internally consistent MLIP gives forces from its energy surface; internal consistency alone does not make that surface correct relative to DFT. Numerical integration accuracy and reference accuracy are separate tests.

Before production, every model passed paired 1 fs and 0.5 fs constant-energy tests from exactly cloned initial states at two compositions and 493.15 K. UMA and MACE passed two further composition-extreme checks. A provisional numerical target limits total-energy range and endpoint drift per atom to 1% of the thermal kinetic-energy scale $dk_BT/(2A)$, where d is the number of physical velocity degrees of freedom. The extreme-case 1 fs ranges were 0.024–0.038 meV/atom and decreased to 0.006–0.011 meV/atom at half timestep. This establishes a bounded numerical test, not general MD stability.

We also ran 50 one-ps NPT pilots. UMA/MACE passed the extended-energy check for every operating point. Production uses 20 distinct mask lineages, five coupled operating points and two models: 200 trajectories. Each has 1 ps thermal preparation followed by 5 ps NPT sampling, at 1 fs. The thermostat/barostat relaxation times are 100/1000 fs. These are exploratory settings and durations.

Temperature (K)	Pressure (MPa)	Purpose
298.15	0.101325	Room-condition reference
393.15	0.08	Project operating point
433.15	0.46	Project operating point
473.15	1.32	Project operating point
493.15	2.02	Project operating point

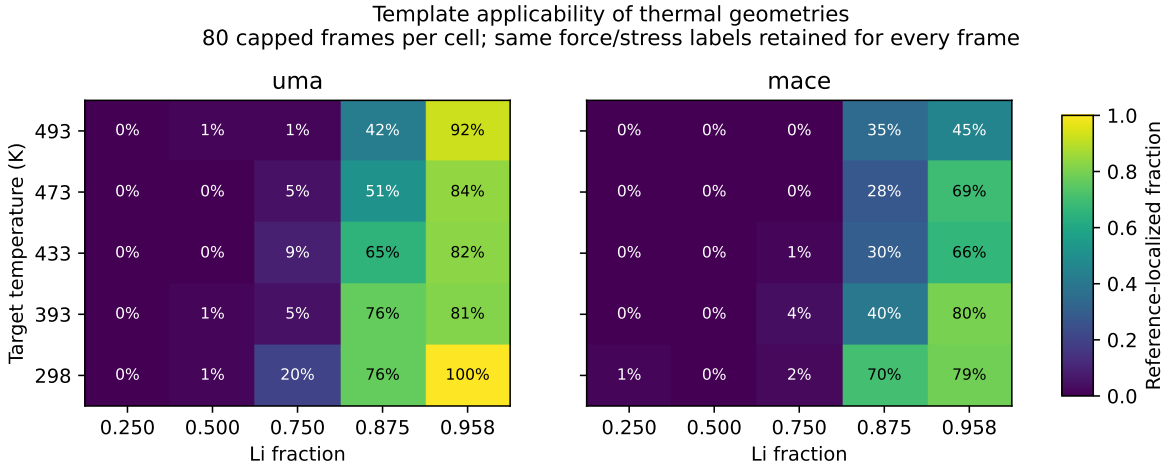
The four project pressures come from actual archived input manifests; the first is 0.08 MPa. Targets are metadata. Measured pressure includes the kinetic stress term; it is never replaced by a target. Pressure conversion is $P_{\mathrm{eV/Å^3}} = P_{\mathrm{MPa}} 10^6/(1.602176634\,10^{11})$. The older TorchSim conversion issue is recorded separately and does not explain every historical MD failure.

Production receipt: 200 terminal complete trajectories; 4000 capped frames from 20 parent lineages. The full traces and physical snapshots are preserved. The final-half trajectory summaries are descriptive, not equilibrium means with established error bars. These five coupled T/P points cannot separately identify independent temperature and pressure effects.

Thermal export removes affine strain and global translation, while keeping raw coordinates unchanged. A periodic one-to-one Li-to-site assignment diagnoses current occupancy. Displacement/force arrays remain in *ancestral* slots; observed occupancy cannot replace their mask without reanchoring rows. Unlocalized frames are flagged rather than called O3 automatically. Thermal frames are not filtered by an optimization force threshold. Physical NPZ states are not exact restarts of the MTK extended integrator state. For thermal conditioning, start with target T/P and the geometry cohort, while keeping instantaneous T/P as measured diagnostics. Near-zero force and stress targets describe minimization, rather than typical finite-temperature frames.

8. Which thermal frames fit a parent-site model?

Numerical completion does not mean a trajectory stays near its starting site pattern. Only 636/2,000 UMA frames and 440/2,000 MACE frames meet the conservative parent-site localization bound. Every frame meeting that bound also retains its ancestral occupancy in the independent one-to-one Li assignment. All 4,000 raw frames, energies, forces and stresses remain available; these flags determine representation applicability, not whether a frame should be erased.



The bound requires each present atom's displacement to be less than half the minimum same-species parent spacing, multiplied by the smallest cell stretch. By the triangle inequality this is a sufficient guarantee that the assigned parent site is its unique nearest same-species site. Failure is inconclusive: it may reflect a Li hop, host rearrangement or a reference that no longer fits. It is not a phase label or a universal physical-disorder threshold.

We inspected 12 trajectories across both models, Li counts 12/24/46 and 298/493 K. Warm-start coordinates exactly matched their immutable relaxed seeds; NPT started exactly at the warm endpoint. No half-cell fractional jumps were observed between saved snapshots in these cases. Independent periodic Co/O registration still left large low-lithiation host displacements. For example, one UMA Li12 trajectory at 298 K went from 0.13 Å host RMS at its seed to 0.80 Å after the fixed-cell warmup and 2.39 Å at the NPT endpoint; the alternative registration gave 2.35 Å. Li46 controls remained much closer.

This supports large *model-predicted internal motion*, often concentrated in plane. It does not identify O1/spinel or establish DFT-level behavior. The motion begins before NPT in some examples, so a barostat-only explanation is insufficient. Small residual force at a relaxed seed establishes approximate stationarity, not stability: near a stationary point the second-order energy change is $\frac{1}{2}\delta r^T K \delta r$, where K is the energy Hessian in eV/Å^2. We have not established its positive curvature or a physical phase transition. Activated motion, soft modes and MLIP error remain possibilities.

Training decision. Keep the initial dense geometry experiment on the packaged UMA fixed-cell cohort. Localized variable-cell or thermal frames can be added with explicit cohort labels, composition coverage and lineage weights. The other thermal frames need a more general structural representation or careful reassignment. If Li rows are reanchored to an observed mask, audit current-mask overlap across splits as well: the original lineage split alone does not prove current-vacancy novelty after hopping.

9. Work division and the next discriminating tests

Ricardo: chemistry, data and physical scoring. Own parent/site definitions, symmetry canonicalization, source Hamiltonian namespaces, split lineage and MLIP/DFT interfaces. Inspect variable-cell structural changes using cell metrics, coordination and site assignments. Define which energy target is being learned. Later DFT confirmation and MLIP fine-tuning depend on a sampler result worth testing; they are not required for the mid-term.

Beihao: conditional generative modeling. Reproduce the frozen mask CVAE, composition-only model and VAE; inspect reconstruction, validation ELBO, KL per latent dimension and three-seed variability. Improve one mechanism at a time: for example, compare the binary class with a train-derived continuous energy condition, or a decoder that represents site correlations more explicitly. Freeze settings and checkpoints before new oracle assessment. The geometry prototype must retain deterministic regression as its control.

Varad: baselines and evaluation. Own equal-attempt and equal-query comparisons, mutation/uniform controls, count validity, symmetry novelty, nearest-training-orbit distance, condition responsiveness, per-composition results and seed plots. Do not pool absolute energies across teachers or compositions. Track offline fitting cost separately from proposal/oracle cost. An energy/novelty result must be traceable to a manifest row and frozen model. The current fresh oracle pool is assessment data for the reported fits; training on it in a later round requires a new assessment pool and novelty cache.

By Nov. 9: produce the approximately five-page mid-term with revised scope, mathematical setup, algorithms, evidence, advantages and limitations. Each collaborator should budget roughly 6–8 focused hours before submission, including a joint integration pass. The existing results are sufficient; new DFT or a universal cross-material generator is not a dependency.

For Nov. 24 / Dec. 1 presentations: prioritize a reproducible sampler comparison, seed-stability explanation and one supported improvement or clear negative result. Allow roughly 4–6 hours per person per week, adjusted to actual availability. Coordinate who presents on the assigned course date. Geometry starting-point acceleration is the strongest extension if the core is stable.

By Dec. 12: complete the final report with experiments and conclusions. A small independently chosen DFT validation set and MLIP fine-tuning comparison can be added after a sampler passes an agreed practical utility test. They should not displace adequate generative-model controls.

Low-energy configurations alone are insufficient training coverage for MD. At a relaxed minimum, nearly zero forces teach stationarity but little about forces on displaced atoms or response to strain. Retain intermediate and thermal frames, and validate force/stress errors and MD behavior separately from vacancy-energy quality.

The next useful tests are specific. First, check whether any model improvement beats mutation while preserving exploration distance under the same query budget; seed averages alone are insufficient. Second, test relaxed-energy conditioning on genuinely paired relaxed labels if relaxed structures are the objective. Third, time initialization, decoding, scoring and required relaxations before claiming computational savings. Additional epochs, larger cells or more labels are not automatically remedies for the observed latent/conditioning problem.

Size/material transfer and reconstructive phases remain later work. A periodic site graph with local stochastic variables can support variable numbers of sites. On a transitive lattice, a shared global latent with otherwise identical node features gives identical logits in the stated factorized Bernoulli decoder. Local noise, autoregressive conditioning or another structured joint decoder can supply ordering correlations. Geometry generation also needs vector/cell transformation rules. The supplied graph smoke test checks this symmetry issue; it is not a trained GNN performance result.

10. Files, reproducibility and limits of the evidence

The portable learning package includes five small teacher-label arrays, the 48-site parent/permutations, frozen splits, the UMA fixed-cell displacement dataset, CPU/CUDA trainers, a fixed-count sampler and four example checkpoints. It requires NumPy/PyTorch and no MLIP installation. Output directories must be new. Two-epoch CPU fits, checkpoint reload and sampling were actually tested; the production fits used the pinned GPU environments. The broader platform should accept a pristine structure and target species (for example Na instead of Li), with site indices and symmetry defined from that input. This pilot's dense network dimensions remain fixed to LCO; variable-size or cross-material generation needs a different representation and validation.

```
cd collaborator_package
python cvae/train.py --teacher uma --kind cvae --device auto \
  --seed 20261106 --output-dir runs/new_cvae
python cvae/portable_sample.py \
  --checkpoint checkpoints/uma_cvae_main_20261106/best.pt \
  --n-li 24 --count 100 --sampling conditional --device cpu \
  --output samples_Li24.npz
```

Campaign path	Meaning
`data/parents/`	Pristine geometry, Li indices, masks, symmetry permutations
`data/learning_*.npz`	Model-independent arrays; teacher-specific labels
`splits/`	Frozen lineage and composition-interpolation splits
`data/geometry_*.npz`	Capped, aligned fixed/variable/thermal frames
`data/generated/`	Raw five-teacher fresh-proposal E/F/stress shards
`cvae/runs/`	Sixteen fitted mask models, histories and configurations
`cvae/geometry_runs/`	CVAE and deterministic geometry prototypes
`reports/`	Figures, attempt-level and grouped metrics
`validation/`	Independent checks and physical/provenance receipts
`resources.json`	Allocation accounting and interruption/recovery record

The aligned relaxation exports contain 54,810 frames, fewer than the raw snapshots. Variable-cell localization is a sufficient geometric check, not a phase classifier: SevenNet has 4,094/5,000 frames within that bound. Preserve the other frames for graph/reassignment analysis rather than training the simple fixed-template model on an assumed unchanged site map.

A separate bounded diagnostic rescored 16 actual archived Hu geometries each from O3, O1 and spinel, including optimized and distorted frames. The exact archived DFT labels and provenance remain attached. Raw energy biases vary by model/reference convention and must not select a winner by themselves. Force and stress errors are tabulated too; this small stratified diagnostic is not a comprehensive phase benchmark. No new DFT labels or recertification of archived electronic/spin convergence is implied.

O3, O1 and spinel are structural families, not vacancy masks. Their coordinate relationships and electronic states can differ. Likewise, different DFT parameters need not produce merely a constant energy shift: that hypothesis requires same-geometry, same-composition testing. All new large-cell results here are MLIP-oracle results. Ground-state discovery, phase stability, equilibrium thermodynamics and improved DFT-level MLIP training remain unestablished.

Primary methodological references. Kingma and Welling, Auto-Encoding Variational Bayes, Sec. 2.2, Eqs. (1)–(3), establishes the variational objective. Sohn, Lee and Yan, Learning Structured Output Representation using Deep Conditional Generative Models, Sec. 3, supplies the conditional formulation. The implemented dynamics and optimizer conventions are traceable to the pinned ASE 3.26 source and saved campaign scripts; the tolerances and short-run gates in this report are our explicit preliminary design choices, not recommendations from those papers.

Resource snapshot. 37.88 allocated four-GPU-node-equivalent hours, against the authorized 200-node-hour cap; GPU-device hours are divided by four. This includes allocated idle/loading time, not just calculator time. This snapshot precedes the final document review; `resources.json` records the final total and terminal allocation states.